Tae Hong (Richard) Kim

richardme.com |th32kim@uwaterloo.ca | +1 (905) 781-8352

Technical Skills

- **Languages:** Java, C#, TypeScript/JavaScript, Python, SQL, C/C++, PowerShell
- **Frameworks & Data:** Spring Boot, ASP.NET Core/.NET, React, Angular, Next.js, PostgreSQL, Oracle DB,
- **Cloud & DevOps:** Azure, AWS, Docker, Kubernetes, Terraform, GitHub Actions, Git, Linux
- **AI & Tools:** LangGraph, LangChain, OpenAI API, GitHub Copilot SDK, Pydantic, Drizzle ORM, SQL Server

Experience

Toronto Dominion Bank (TD)
Sept 2026 – Dec 2026

Software Engineering Intern
Toronto, ON, Canada

- Developed a **C#/ASP.NET Core backend service** and **Angular dashboard** to process system change requests, apply application-specific validation rules, and integrate the **GitHub Copilot SDK** for AI-assisted validation with structured JSON responses, reducing manual document review by **60%**.
- Built an automated **backend validation workflow** across TD Wealth systems using **PowerShell, SQL Server,** and **AutoSys** to detect inactive user accounts before downstream **API** execution, reducing manual verification and failed processing.
- Supporting production delivery of **C#/ASP.NET Core** and **Angular applications** through **GitHub Actions, SonarQube, Nexus,** and **Digital.ai**, automating build, testing, quality-gate, artifact publishing, and deployment workflows targeting an **80%** *reduction* in manual deployment effort.

FundServ Inc
Jan 2026 – April 2026

Software Developer Intern
Toronto, ON, Canada

- Built and productionized secure **B2B microservices** deployed to **Azure Kubernetes Service (AKS)** using **Azure APIM, Terraform,** and **Helm**, implementing **MSAL, PKCE,** and **JWT authentication** for secure communication between applications.
- Automated **CI/CD** and **software delivery workflows** using **GitHub Actions, Docker, Azure Container Registry, ArgoCD, Veracode, Vault,** and **JaCoCo**, reducing release effort by **70%** while enforcing security scanning and **80% unit-test coverage** before production deployment.

FundServ Inc
Jan 2025 – April 2025

Software Developer Intern
Toronto, ON, Canada

- Built a **full-stack onboarding microservice** using **Java, Spring Boot, TypeScript, Angular, and Oracle DB**, implementing backend services and frontend workflows to automate financial-institution onboarding processes.
- Developed a **backend CSV ingestion and data validation service** using **Spring JPA, SQL, and Unix file systems**, reducing onboarding errors by **40%**.
- Designed end-to-end application workflows and produced **HLD/LLD system architecture documentation** for technical design reviews, supporting CIO approval for production deployment.

Ford Pro
Sep 2023 – Dec 2023 | May 2024 – Aug 2024

Exploratory Test Engineering Intern
Oakville, ON, Canada

- Developed automated **API, database,** and **end-to-end validation workflows** using **Playwright, Postman, SoapUI,** and **SQL**, accelerating engineering feedback loops by **50%** while monitoring production services with **Datadog**.
- Improved **CI/CD build and deployment workflows** using **Tekton, Docker, Azure DevOps, Linux,** and **shell scripting**, applying job parallelization and caching to improve build efficiency and deployment reliability for distributed services.

Projects

AI Website Generator - Next.js, TypeScript, OpenRouter, Drizzle ORM, PostgreSQL
Th32kim/AIwebGen

- Built a **full-stack AI application** using **Next.js, TypeScript, and OpenRouter APIs** that converts natural-language prompts into generated user interfaces and streams LLM output through **Server-Sent Events (SSE)** into sandboxed previews.
- Developed a type-safe **backend persistence and versioning layer** using **Drizzle ORM and Neon PostgreSQL**, enabling users to store, retrieve, and iterate on AI-generated interface configurations with **sub-second rendering**.

AI-Agent Search Engine - Python, LangGraph, LangChain, OpenAI API, Pydantic
Th32kim/AI-Agent

- Built a **Python AI research agent** using **LangGraph** to orchestrate Google, Bing, and Reddit retrieval through asynchronous **Bright Data** ingestion and snapshot polling, separating long-running data collection from LLM requests.
- Implemented multi-stage **LangChain** and **OpenAI reasoning** over retrieved data using **Pydantic structured outputs** for validated URL selection, targeted Reddit retrieval, and synthesis of multi-source results into a unified response.

Education

University of Waterloo
Sep 2022 – May 2027

Bachelor of Applied Sciences - Computer Engineering Honours Co-op
Waterloo, Ontario, Canada

Courses: **Computational Intelligence** (**ML**), **Database Systems** (**SQL**), **Systems Programming and Concurrency (C)**